

A Model of An Electron Includes Two Perfect Black Bodies

Satoshi Hanamura*

This paper modifies two significant points of the existing idea theory. First, replaced the image of a virtual photon ~~is~~ replaced with a real one. A bare electron captures the photon. Second, ~~but a new assumption~~ it is assumed that the bare electron is a perfect black body. To meet the constraints of charge conservation, a the virtual photon ~~has to~~ must include two bare electrons. The concept of the black body needs to applying thermodynamics. There is ~~the~~ a temperature gradient between the two because the two particles have a role of emitter alternate between behaving as emitters and absorber one after the other absorbers. That is, Presently, it is considered an electron, what we called would be consisted of, consists of two blinking two bare electrons ~~with~~ and at least one real photon.

I. INTRODUCTION

Regardless of vigorous exertion In spite of significant experimental efforts, a perfect black body [H][1] has not yet been found yet [2] in nature [2]. In This paper study, not to only provide a proof proves that an electron is a perfect black body. Instead, set a principle, but assumes that the electron this is the one. By doing so, that principle can case and build upon this assumption to contribute to making a flourishing aspects branch of physics. It is natural to think that an electron is not a perfect black body. In spite of that, remake; nevertheless, remaking our view of the particle and rebuilt the concept of electron brings other images of oscillators of electron allows us to imagine this elementary particle, as consisting of oscillators.

In the beginning, the modeling deal with the I begin by discussing virtual photons and fluid dynamics. Disease, upon which the proposed model is based. Then, the structure of the Sun is discussed and it is assumed that an electron has similarities on the point in terms of the being a perfect fluid. That (a concept that comes astrophysics from astrophysics and fluid dynamics). These fields of knowledge provide form the center of the proposed model to the centering force.

On the other hand contrary, this electron model dig into the also suggests a structure of an for the electron. Assume It is assumed that an electron consists of two bare electrons and with both of them as being black bodies. This reorganization taps the constitutes a new image of an the electron.

Next, these These two bare electrons exchange their heat energy through the virtual photon. The way of transmitting energy is in the form of radiation. It is through virtual photons. This is possible to be the way because those two bare electrons are inside of the virtual photon. The two black bodies exchange its heat energy through the virtual photon as a real photon.

During While the two black bodies of emitting emit and absorbing its absorb heat energy, they do not have equal value of emitting and absorbing due to do so in an equal proportion because of a margin cosine and sine powered four four. The total energy is unchangeable does not change because this model uses an isolated system. For keeping To ensure that energy conservation law is conserved, the rest of amount shall be transported by the virtual photon as in the form kinetic energy.

We It will find be shown that the virtual photon is moving moves in space with its velocity changing from zero to maximum in keeping with a manner similar to a harmonic oscillator. The velocity comes from energy caused released [2.1 [JP] 英単語・語句のご確認をお願いします。] by the velocity potential. The temperature gradient makes forms this potential. The temperature This gradient, is composed of in turn, arises from the energy difference of between the two bare electrons.

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コメントの追加 [Editor8]: Remark: Earlier you claimed that they were inside the electron. Please clarify this relationship.

コメントの追加 [Editor9]: Remark: Note that this is unclear and seems to contradict the idea that the photon is virtual. Please clarify.

コメントの追加 [Editor10]: Remark: This is unclear. What does this sinusoidal function describe? Please clarify.

コメントの追加 [Editor11]: Remark: To what amount are you referring? Please clarify.

For reference, this model ~~concerned the concerns electrons moving~~ significantly below the speed of ~~the electron is too slow~~ compared to the velocity of light. That is, there is no effect of the relativity theory in this paper; thus, relativistic effects are not considered.

II. MODELING

A. ~~Modify~~Modification of a Virtual Photon

Image a free electron. For ~~making at the purposes of this~~ electron model, change the concept of a virtual photon into a real one: ~~in~~ this paper, the word "a term "virtual photon-" shall be used ~~asto describe~~ a real photon ~~which captured by~~ a bare electron with coupling constant, ~~a of a~~.

To ~~make an clarify our~~ image of ~~a captured a~~ virtual photon ~~clear~~, we use ~~a the~~ concept of "hydrostatic equilibrium" ~~in~~ from fluid mechanics.

Gravity is balanced by a pressure-gradient force in the Sun. ~~An The~~ inner structure of an electron is comparable ~~towith~~ the balancing ~~in of~~ the Sun as ~~a~~ self-gravitating gas sphere.

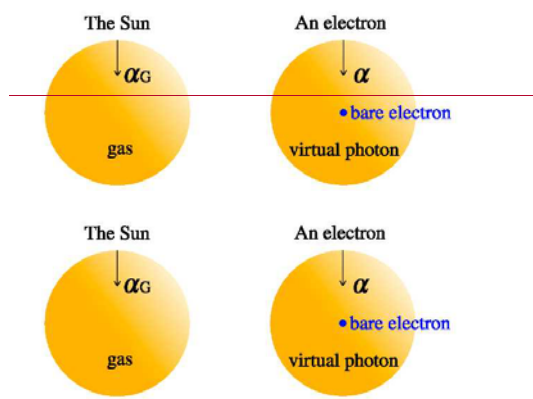


FIG. 1. Two spheres and the central force

Gravity ~~comes from points toward~~ [2,1 [JPI 英単語・語句のご確認をお願いします。] the center of the Sun. ~~That is equivalent to~~. Equivalently, the electromagnetic force ~~comes from points toward~~ the center of an electron.

~~It is well established that the way~~ A means of determining the internal structures of stars. ~~Calculate the is well established~~. First, ~~calculate the stellar~~ radius ~~of the stars by~~ using the hydrostatic equilibrium equation [5] ~~in the field of~~ [5] from fluid dynamics:

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} \quad (11.1)$$

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コメントの追加 [Editor13]: Remark: We infer that you mean that the photon was captured by the electron. Please check whether this is the case.

コメントの追加 [Editor14]: Remark: Note that Figure 1 has not been cited in the main text. Please check.

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} \tag{II.1}$$

This equation ~~derives~~yields the Lane—Emden equation, Eq. ~~(II.2)~~ (II.2):

$$\frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -G \frac{dM_r}{dr} \quad (\text{II.2})$$

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The pressure P ~~connected~~is related to the density ~~pas~~ P:

$$P = K \rho^{1+\frac{1}{n}} \quad (\text{II.3})$$

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~~This is the solution~~ known as the polytropic equation, ~~W~~where K is constant. The last two equations ~~can belong~~belong to the field of astrophysics.

~~Polytrope is~~Polytropes are one ~~of the kind~~type of fluid. Using the ~~poly-tropie~~polytropic equation, ~~such polytropes~~ can be regarded ~~polytrope as a~~perfect fluid. ~~Apply~~fluids. Let us apply this fact to the virtual photon in this model.

However, referencing Coulomb potential; $V = (-e^2)/(4\pi\epsilon_0 r)$, the ~~spherically symmetric~~ potential energy ~~is spherically symmetric~~ depends only upon the distance, r. In ~~like a~~ similar manner, let us hypothesize that the shape of the virtual photon is spherical.

To ~~get~~simplify the model ~~simple~~, we ~~set an assumption as follows~~assume the following.

Assumption: A virtual photon is ~~the~~a perfect fluid and captured by ~~a~~the central force of a bare electron.

Therefore, the virtual photon captured by ~~electromagnetic~~the electromagnetic force ~~doesn't~~does not emit any electromagnetic radiation and has no oscillation at this moment. By this logic, a virtual photon can be replaced by a real photon spread over a certain length. ~~Yet~~; a bare electron has ~~<1018 m~~a radius ~~<1018 m~~.

~~Modify~~Thus, let us modify the term "virtual photon" to have meaning ofrefer to a real photon captured by the bare electron, which exists all the time.

Assumption: Reinterpret a virtual photon as a real photon ~~which is~~captured by the potential energy ~~driven by~~of a bare electron.

~~For making~~To clarify this free-electron model ~~clear~~, ~~discuss~~, I explain that only the bare electron has ~~the~~potential energy in ~~subsequent~~a subsequent section. That is, the virtual photon has ~~whole~~all kinetic energy and the bare electron has all potential energy.

B. ~~An~~Electron Contains Black Body~~ies~~

In this subsection, I interpret the bare electron as a black body. ~~Making~~To push the model ~~push~~forward, ~~set more an as~~sumption~~let us make another assumption~~.

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Assumption: A bare electron is a perfect black-body and radiates its energy ~~harmonically by~~harmonically through time-dependent oscillation.

This bare electron has a bare mass, ~~and we handle its mass by temperature, T. Postulate- We postulate that~~ this black body oscillates time-dependently, ~~by- according to an arbitrarily chosen cosine oscillator made an arbitrary assumption function:~~

$$\begin{aligned} E_{\text{bare}} &= T \cos^4 \left(-\frac{\omega t}{2} \right) \\ &= T \cos^4 \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.4})$$

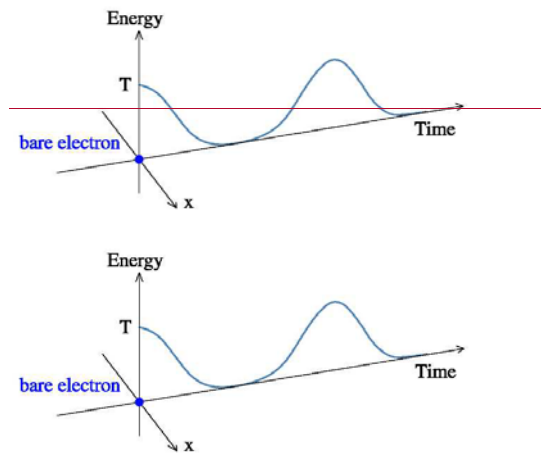
$$\begin{aligned} E_{\text{bare}} &= T \cos^4 \left(-\frac{\omega t}{2} \right) \\ &= T \cos^4 \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.4})$$

There is no element $\cos(kx)$ in the above equation. ~~The reason is, because~~ we assumed the bare electron ~~has to have a~~ fixed position and ~~doesn't not to~~ oscillate back and forth in space. This immovability is shown by using the delta function:

$$E_{\text{bare}} = \int_{\text{all space}} T \cos^4 \left(\frac{\omega t}{2} \right) \delta^3(\mathbf{r} - \mathbf{a}) dx dy dz \quad (\text{II.5})$$

$$E_{\text{bare}} = \int_{\text{all space}} T \cos^4 \left(\frac{\omega t}{2} \right) \delta^3(\mathbf{r} - \mathbf{a}) dx dy dz \quad (\text{II.5})$$

To ~~makesimplify~~ the model ~~simple, take, assume~~ one-dimensional coordinates and set the center of ~~the~~ bare electron ~~on the~~ point ~~at~~ ($x = 0$). The change of energy of the bare electron with respect to time ~~as is~~ shown in Fig.-2.



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FIG.-2. Plots of $T\cos 4(\theta)$, which is the change ~~of in the~~ energy waveform with a bare electron at location $x = 0$.

Figure 2 shows that ~~at the~~ bare electron ~~as~~ black body has two phases. One is ~~a phase of an absorber absorbing~~ and the other is ~~an emitter of radiation emitting~~.

Using 8 instead of θ , ~~classify~~, $\cos 4(\theta)$ is classified into two phases by reference to Fig.-3. As a bare electron is a spinor, ~~we~~ ~~can see~~ the same curve is observed when plotting $f(8) = \cos 4(\theta)$ instead of $T\cos 4(\theta)$ or $T\cos 4(\theta^*)$ with $T = 1$.

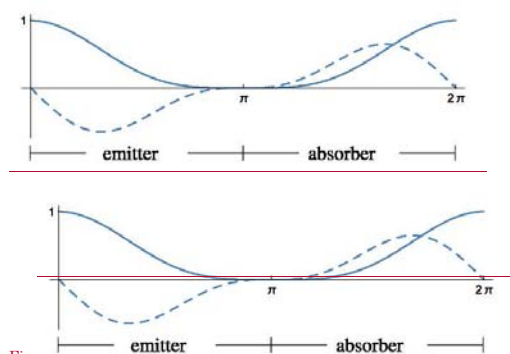


Fig.

FIG. 3. Plots of $f(\theta) = \cos^4(\frac{\theta}{2})$ and $f'(\theta) = -2 \cos^3(\frac{\theta}{2}) \sin(\frac{\theta}{2})$

Except stationary points on $f(\theta) = \cos^4(\frac{\theta}{2})$, if $f'(\theta) = -2 \cos^3(\frac{\theta}{2}) \sin(\frac{\theta}{2})$ were minus, while $0 < \frac{\theta}{2} < \pi$ in Fig. 3,

the energy of the ϕ . Plots of $f(\theta) = \cos 4\left(\frac{\theta}{2}\right)$ and $f'(\theta) = -2\cos 3\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right)$

Except stationary points on $f(\theta) = \cos 4\left(\frac{\theta}{2}\right)$, if $f'(\theta) = -2\cos 3\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right)$ were minus, while $0 < \frac{\theta}{2} < \pi$ in Fig. 3, the energy of the black body is decreasing. This is the emittance phase is emitter. On the other hand, if $f'(\theta) = -2\cos 3\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right)$ were plus positive, the energy of the one [1.2 [JP] 文章が不明瞭です。意味を明確にしてください。] is increasing due to energy absorption.

C. The Necessity of the Paring Oscillator

According to Fig. 2 and Fig. 3, the bare electron energy hits zero when $\theta = \pi$. If the bare electron vanished at $\theta = \pi$, this allows the virtual photon without a bare electron to exist. That contradicts the contradicting [2.1 [JP] 英語・語句のご確認をお願いします。] charge conservation, since photons cannot carry any electric charge.

To keep the maintain charge conservation, we take in another bare electron. These two electrons play a role of act as both emitters and absorbers in turn.

To meet the requirements for working of emitter and absorber simultaneously simultaneous emission and absorption, assign different phases to each other. Set the two electrons as paring oscillator to paired oscillators with $Te1 = \cos 4(\phi)$ and $Te2 = \sin 4(\phi)$. Plug 1 into Te, and take a look at Fig. 4.

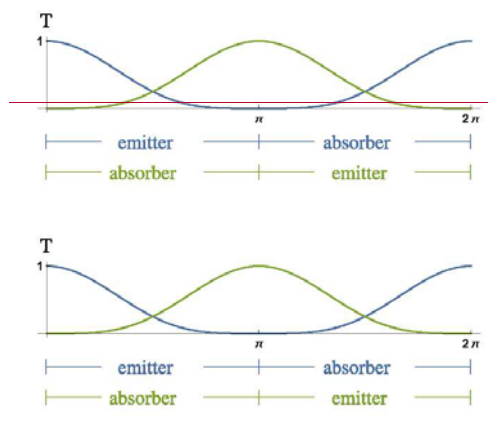


FIG. 4. Plots of $Te1 = \cos 4(\phi)$ in (blue line) and $Te2 = \sin 4(\phi)$ in (green line).

While $0 < \phi < \pi$ in Fig. 4, one of the bare electrons, $Te1$, is in the emitter phase. On the other hand, $Te2$ is in the absorber phase. $Te1$ emit its energy by radiation. The more $Te2$ absorb the energy of $Te1$ emits energy by radiation, which is in turn absorbed by $Te2$.

~~Besides~~Moreover, these radiation exchanges can occur within the virtual photon. Radiated energy propagates via ~~elec-~~romagnetic wave. ~~This fact reinforces the~~electromagnetic waves, ~~thereby reinforcing these~~ exchanges.

D.— The Interval between the Two Bare Electrons

~~Another bare electron would emerge in the vicinity of the virtual photon. Because of exerting, because the two bare electrons exert~~ repulsive ~~force on two bare electrons, forces.~~ Both of the two bare electrons consist of an electron. (2.1 [JP] 英単語・語句のご確認をお願いします。)

~~For making~~To simplify the model ~~simple, regarding the shape of, regard~~ the virtual photon as a sphere. Furthermore, ~~make a restriction that a restrict the~~ bare electron as ~~a~~electrons, which are condensed matter of heat energy ~~appears, to appear~~ and disappears at the same point. ~~These are for making this model simple.~~

Assumption: Another pair of ~~the~~ bare electrons will emerge on the sphere ~~at~~ the virtual photon's surface.

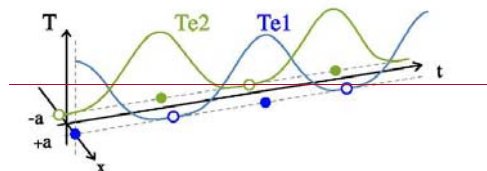
This assumption works ~~for keeping to maintain~~ the "P" principle of locality. Another pair of ~~the electron~~electrons would not emerge ~~abruptly, for example,~~ on the far side of the moon ~~abruptly~~. This postulation implies that the information of the two bare electrons ~~are is~~ shared within ~~a~~the virtual photon.

~~For investigating~~To investigate the relationship between the photon and the two bare electrons, divide ~~a~~the bare electron T_e in Eq. (II.5) into (II.6) and (II.7). With an Eq. (II.6), with initial value $x = -a$,

~~$$T_{e1} = \int_{\text{all space}} T_e \cos^4 \left(\frac{\omega t}{2} \right) \delta(x - a) dx \quad (\text{II.6})$$~~

~~Set another paring oscillator T_{e2} and Eq. (II.7), with an initial value $x = -a$ as well, $x = -a$.~~

~~$$T_{e2} = \int_{\text{all space}} T_e \sin^4 \left(\frac{\omega t}{2} \right) \delta(x + a) dx \quad (\text{II.7})$$~~



$$T_{e1} = \int_{\text{all space}} T_e \cos^4 \left(\frac{\omega t}{2} \right) \delta(x - a) dx \quad (\text{II.6})$$

$$T_{e2} = \int_{\text{all space}} T_e \sin^4 \left(\frac{\omega t}{2} \right) \delta(x + a) dx \quad (\text{II.7})$$

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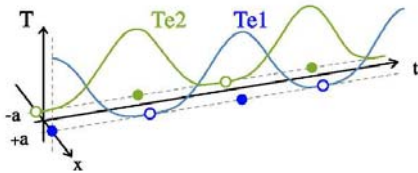


FIG.-5. Plots of the bare electrons apart from ~~each other~~ one another.

Fig.-5 is drawn in Figure 5 is a three-dimensional perspective based on the Fig.-4. The light green circle ~~on~~ at $x = -a$ is drawn with indicates the outer frame. That means ~~that~~ the paring oscillator, Te2, does not have any heat energy at $t = 0$. Te1 has ~~whole~~ all heat energy in the system at $t = 0$.

E.— Applying the First Law of Thermodynamics

~~"We use the first law of thermodynamics as a special case for considering internal energy: "An isolated system does not allow for passage of matter or energy into or out of the system" [3]. [3]. And that is a special case for considering internal energy.~~

~~For~~ ~~us~~ ~~Let us focus~~ our attention on an electron model composed of one virtual photon and two spinors of bare electrons as an "isolated system". That is, the electron does not allow the spinors to ~~go out anywhere from the photon~~ leave or to pass ~~energy~~ outside ~~the energy~~ of the photon. ~~Review the law,~~

"The first law of thermodynamics: For an isolated system, the total energy of the system remains constant" [3].

The law is expressed in equation by using the total amounts of work, w , and heat, q :

$$U = q + w \quad (11.8)$$

$$U = q + w \quad (\text{II.8})$$

"If energy cannot move in or out, then the total energy U of the system does not change" [3]. This isolated system is "adiabatic" with respect to outside the surroundings. However, the inside is not adiabatic. Because the, because two differential-phase spinors are in the system present. In this situation, $q = 0$. Apply Let us apply the isolated system assumption to this electron model, heat, Eq. (II.8). Heat, q , is equal to $T_{e1} + T_{e2}$, giving us

$$U = T_{e1} + T_{e2} + w \quad (\text{II.9})$$

$$U = T_{e1} + T_{e2} + w \quad (\text{II.9})$$

However, this model does not deal with the pair production from two-photon interaction. The virtual photon does not vanish by with the passage of time passing. Same as the role of allotting kinetic energy. Moreover, the bare electron does not have kinetic energy, but the heat energy.

In these requirements, making the roles of these particles separately [2.1 JPN 英単語・語句のご確認をお願いします]. Bare electrons take the absorb heat energy and, whereas the photon carries the kinetic energy. Apply Let us apply these roles to Eq. (II.8), (II.9); as the kinetic energy of the virtual photon, we have

$7k \cdot e^{\wedge}$

$$\Delta U = \Delta T_{e1} + \Delta T_{e2} + \Delta \gamma_{K.E.}^* \quad (\text{II.10})$$

$$\Delta U = \Delta T_{e1} + \Delta T_{e2} + \Delta \gamma_{K.E.}^* \quad (\text{II.10})$$

For an isolated system, $\Delta U = 0$,

$$0 = \Delta T_{e1} + \Delta T_{e2} + \Delta \gamma_{K.E.}^* \quad (\text{II.11})$$

$$0 = \Delta T_{e1} + \Delta T_{e2} + \Delta \gamma_{K.E.}^* \quad (\text{II.11})$$

Hence,

$$\frac{dT_{e1}}{dt} + \frac{dT_{e2}}{dt} + \frac{d\gamma_{K.E.}^*}{dt} = 0 \quad (\text{II.12})$$

$$\text{That } \frac{dT_{e1}}{dt} + \frac{dT_{e2}}{dt} + \frac{d\gamma_{K.E.}^*}{dt} = 0 \quad (\text{II.12})$$

which shows that energy conservation is conserved in this model of the electron of this model.

F. — Fourier's Law

In this section, look through we examine how the virtual photon is captured by the bare electron to move and work in the isolated system. Having as Our expected image is that of a simple harmonic motion is our expectation oscillator.

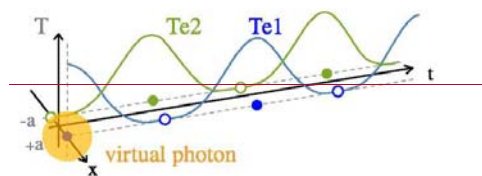
Before looking into the motion by phase, let us check the two fixed states of the positional relationship between the virtual photon and a pair of two oscillators.

These two states have two phases. One comes from $\cos^4(\varphi) = 1$ and the other comes from $\sin^4(\varphi) = 1$.

On the phase of $\cos^4(2\varphi) = 1$, for example at $t = 0$, the center of the virtual photon stays at $x = -a$ in Fig. 6.

According to the Using Fourier's law as a rule of thumb, the heat flux density (units of $\text{W} \cdot \text{m}^{-2}$), q , the material's conductivity (units of $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$), k , and the temperature gradient (units of $\text{K} \cdot \text{m}^{-1}$), T , are related as

$$q = -\kappa \text{grad } T \quad (\text{W} \cdot \text{m}^{-2}) \quad (\text{H.13})$$



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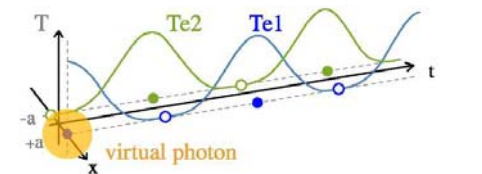


FIG. 6. The position of the virtual photon at $t = 0$. Two point-bare electrons, at two points, $-a$ and a , are separated by a distance of $2a$.

Calculate the temperature gradient between the two bare electrons, is calculated as

$$\begin{aligned} q_e &= -\kappa_e \text{grad } T_e \\ &= -\kappa_e \text{grad } (T_{e2} - T_{e1}) \end{aligned} \quad (\text{H.14})$$

$$\begin{aligned} q_e &= -\kappa_e \text{grad } T_e \\ &= -\kappa_e \text{grad } (T_{e2} - T_{e1}) \end{aligned} \quad (\text{H.14})$$

Replace T_{e1} and T_{e2} in Eq. (H.14) with the harmonic oscillators shown below:

$$T_{e1} = T_e \cos^4\left(\frac{\omega t}{2}\right) \quad (\text{H.15})$$

$$T_{e2} = T_e \sin^4\left(\frac{\omega t}{2}\right) \quad (\text{H.16})$$

$$T_{e1} = T_e \cos^4 \left(\frac{\omega t}{2} \right) \quad (\text{II.15})$$

$$T_{e2} = T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{II.16})$$

The ~~grad Te1 and the grad Te2 have~~include only a ~~time-derivative term. Space-derivative terms of these two; their space derivatives~~ are zero. ~~Because these two, because the bare electrons~~ do not change ~~its~~in position ~~by~~with time. That is,

$$\begin{aligned} \text{grad } T_{e1} &= \frac{d}{dt} \left(T_e \cos^4 \left(\frac{\omega t}{2} \right) \right) \\ &= -2T_e \omega \cos^3 \left(\frac{\omega t}{2} \right) \sin \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.17})$$

$$\begin{aligned} \text{grad } T_{e2} &= \frac{d}{dt} \left(T_e \sin^4 \left(\frac{\omega t}{2} \right) \right) \\ &= 2T_e \omega \cos \left(\frac{\omega t}{2} \right) \sin^3 \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.18})$$

$$\begin{aligned} \text{grad } T_{e1} &= \frac{d}{dt} \left(T_e \cos^4 \left(\frac{\omega t}{2} \right) \right) \\ &= -2T_e \omega \cos^3 \left(\frac{\omega t}{2} \right) \sin \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.17})$$

$$\begin{aligned} \text{grad } T_{e2} &= \frac{d}{dt} \left(T_e \sin^4 \left(\frac{\omega t}{2} \right) \right) \\ &= 2T_e \omega \cos \left(\frac{\omega t}{2} \right) \sin^3 \left(\frac{\omega t}{2} \right) \end{aligned} \quad (\text{II.18})$$

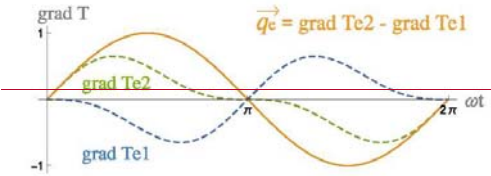
So,

$$\begin{aligned} \text{grad } (T_{e2} - T_{e1}) &= 2T_e \omega \cos \left(\frac{\omega t}{2} \right) \sin^3 \left(\frac{\omega t}{2} \right) \\ &\quad + 2T_e \omega \cos^3 \left(\frac{\omega t}{2} \right) \sin \left(\frac{\omega t}{2} \right) \\ &= 2T_e \omega \cos \left(\frac{\omega t}{2} \right) \sin \left(\frac{\omega t}{2} \right) \\ &= T_e \omega \sin (\omega t) \end{aligned}$$

$$\begin{aligned}\text{grad}(T_{e2} - T_{e1}) &= 2T_e\omega \cos\left(\frac{\omega t}{2}\right) \sin^3\left(\frac{\omega t}{2}\right) \\ &\quad + 2T_e\omega \cos^3\left(\frac{\omega t}{2}\right) \sin\left(\frac{\omega t}{2}\right) \\ &= 2T_e\omega \cos\left(\frac{\omega t}{2}\right) \sin\left(\frac{\omega t}{2}\right) \\ &= T_e\omega \sin(\omega t)\end{aligned}$$

Therefore, put the above conclusion into Eq. (II.14), then calculated (II.14) and calculate qe to below:

~~$$\vec{q_e} = -T_e\kappa_e\omega \sin(\omega t) \tag{II.19}$$~~



~~$$\vec{q_e} = -T_e\kappa_e\omega \sin(\omega t) \tag{II.19}$$~~

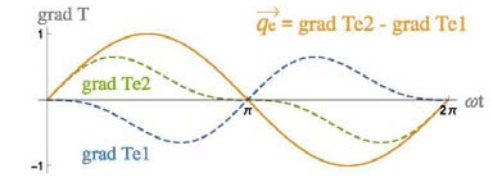


FIG.-7. Plots of qe and two temperature gradients of the spinors, replacing Teui on Eq. (II.17) in Eqs (II.17) and Eq. (II.18) (II.18) with 1 and; Teneus on Eq. (II.19) in Eq. (II.19) with 1.

G. —Velocity ~~p~~Potential and the Virtual Photon

However,Let us focus on the virtual photon. For that, back to the To this end, we return to our assumption of hydrostatic equilibrium.

For preceedingTo advance this model, try to put the virtual photon on a velocity potential [2.1 [JPI 英単語・語句のご確認をお願いします。] To, Two conditions should be satisfied to use the velocity potential, there are two conditions that should be satisfied with-. First, itsthe flow of the fluid should be irrotational. Second, the fluid should be a perfect fluid.

CheckLet us check the first condition-of irrotational. The; the virtual photon moves along with onlythe x-axis, so;

~~$$\omega_{\text{vorticity}} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \tag{II.20}$$~~

Or,

$$\omega_{\text{vorticity}} = \text{rot}(\text{grad}(T_{e2} - T_{e1})) = 0 \quad (\text{II.21})$$

$$\omega_{\text{vorticity}} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad (\text{II.20})$$

That means

or

$$\omega_{\text{vorticity}} = \text{rot}(\text{grad}(T_{e2} - T_{e1})) = 0 \quad (\text{II.21})$$

meaning that a specific velocity potential exists ~~due to~~because of the vorticity ~~is~~being zero.

CheckLet us now check the second condition. ~~For proceeding the modeling~~Previously, the virtual photon ~~would behas been~~ regarded as a polytrope. ~~As for this presupposition, the virtual photon~~Thus, it can be treated as a perfect fluid.

Thus, there is a room ~~for being able to~~ put the virtual photon ~~on~~into the temperature gradient between Te1 and Te2.

Until now, ~~discussed Previously~~ the bare electron ~~has been discussed in the~~terms of heat conduction. ~~From now, proceed modelling of~~conduction. Henceforth, the virtual ~~photon on the~~photon will be modeled in terms of velocity potential.

As the flow met the two conditions, ready for setting new velocity potential for the virtual photon [2.1 [JPL 英単語・語句のご確

認をお願いします。]Generate. Let us generate a new constant, k_y^* (units of $\text{m}^{-1} \cdot \text{s}^{-1} \cdot \text{K}^{-1}$) ~~with~~, satisfying the below condition:

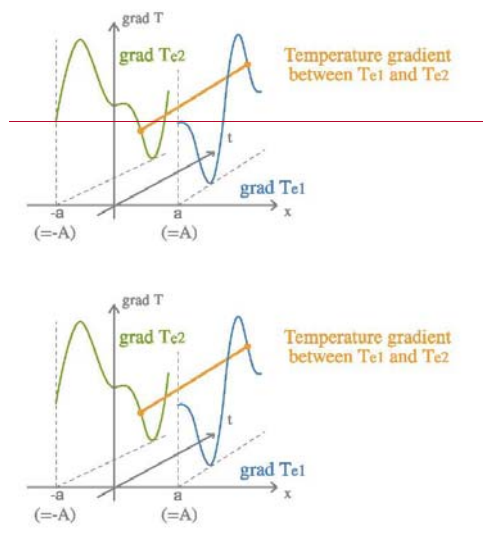


FIG.-8. Relationship ~~of~~between the temperature gradients and the bare electrons in three dimensions.

$$\frac{q_e}{\kappa_e} = \frac{q_{\gamma*}}{\kappa_{\gamma*}} \quad (K) \quad (II.22)$$

$$\frac{q_e}{\kappa_e} = \frac{q_{\gamma*}}{\kappa_{\gamma*}} \quad (K) \quad (II.22)$$

The heat flux, q_e (units of $W \cdot m^{-1}$) comes, arises from the temperature gradient and the speed of a running fluid $q_{\gamma*}$ (units of $m \cdot s^{-1}$) is based on the velocity potential. κ_e is the material's conductivity (units of $W \cdot m^{-1} \cdot K^{-1}$) for the bare electrons. That is the reason why for generating the two different constants, κ_e and $\kappa_{\gamma*}$, are generated.

A governing equation rules Eq. (II.22). Which (II.22), which is to say; the Poisson's equation (kV = -q) provides the fundamentals for describing both q_e and

$q_{\gamma*}$ as the governing equation. The scalar field ϕ , of both the two fluxes is temperature. The source is the bare emitter bare-electron-as, which is a perfect black body.

Assume that the virtual photon moves along with the vector field $q_{\gamma*}$. In other words, the central force composed by arising from the two bare electrons moves along with the velocity potential. Use Eq. (II.19) Using Eqs (II.19) and (II.22), we obtain

$$\begin{aligned} q_{\gamma*} &= \frac{\kappa_{\gamma*}}{\kappa_e} q_e \\ &= -\frac{\kappa_{\gamma*}}{\kappa_e} \kappa_e \text{grad } T \\ &= -\kappa_{\gamma*} \text{grad } (T_{e2} - T_{e1}) \\ &= -T_e \kappa_{\gamma*} \omega \sin(\omega t) \end{aligned} \quad (II.23)$$

$$\begin{aligned} q_{\gamma*} &= \frac{\kappa_{\gamma*}}{\kappa_e} q_e \\ &= -\frac{\kappa_{\gamma*}}{\kappa_e} \kappa_e \text{grad } T \\ &= -\kappa_{\gamma*} \text{grad } (T_{e2} - T_{e1}) \\ &= -T_e \kappa_{\gamma*} \omega \sin(\omega t) \end{aligned} \quad (II.23)$$

In here,

Here, we add a new assumption,

Assumption: The central force belongs to the two bare electrons moves along with the temperature gradient.

Based on the basis of the above principle, the virtual photon captured by the central force of the electron moves along with the temperature gradient as well.

Eq. (II.23) Equation (II.23) shows that the temperature gradient between $\text{grad } T_{e1}$ and $\text{grad } T_{e2}$ produces a force F .

The force drives the velocity of the virtual photon along with simple harmonic motion. Based on the basis of the principle mentioned

above, the virtual photon swings back and forth spatially between the two bare electrons.

Meanwhile, by replacing $TeKY^*$ with spatial amplitude A , Eq. (H.23) is,

$$q_{\gamma*} = A\omega \sin(\omega t) \quad (H.24) \quad q_{\gamma*} = -A\omega \sin(\omega t) \quad (H.24)$$

Integrating the function Eq. (H.24) by with respect to time, we obtain

$$\int q_{\gamma*} dt = A \cos(\omega t) \quad (H.25)$$

By the setting principle as mentioned above that " $\int q_{\gamma*} dt = A \cos(\omega t)$ " (H.25)

Assuming that the virtual photon moves along with the temperature gradient" leads to gradient yields the behavior of the photon.

Eq. (H.25) implies that the displacement of the photon has a wave moving continuously along the x-axis continuously. Compared to the way of. Compared with the motion of the bare electron, the itself, its energy spot of bare electron moves discretely from $+a$ to $-a$.

Take a look at the Fig.-9. This is added the motion of virtual photon along with Eq. (H.25) and rewrote the Fig. 5 Figs 5 and the Fig. 6-6 (2.1 JJP 英単語・語句のご確認をお願いします.). The radius of the virtual photon can be imagined as equal to the being double the width of the spatial amplitude, $2A$. That is length is equal to $2a$ in the x-axis.

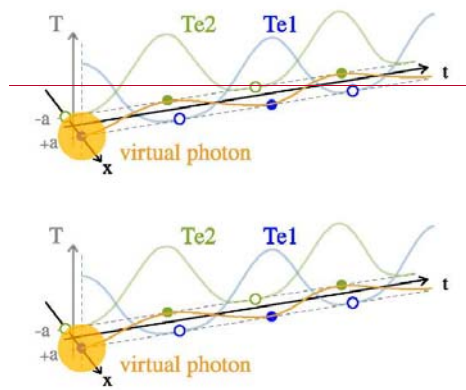


FIG.-9. Behavior of the virtual photon as a spatial simple harmonic oscillator during while the two bare electrons act the role of emitters as emitters and absorbers.

Remarkably, dug out two kinds of vector-vectors, q_e and qY^* , can be recovered from the same scalar field. q_e shows the temperature gradient consisted of the two bare electrons, and qY^* indicates the velocity and spatial value of the virtual photon.

III. DISCUSSIONS

A. —Value of ~~the~~ Emissivity

In the previous section, ~~set the principle it was assumed~~ that a bare electron acts as a perfect black body. ~~Due to the~~ Because of this principle, ~~the virtual photon behaves~~ photons behave like a spatial simple harmonic ~~oscillator Fig. 7, 8 and 9~~ oscillators (Figs 7–9). What happens if ~~its~~ emissivity is less than 1? The temperature of the bare electron ~~is decreased~~ decreases with ~~every each repeated~~ absorption repeated. C. Consider there are two ~~times of the~~ emittance and absorption cycles per 2π , with ~~the~~ frequency f ,

$$\begin{aligned} C &= 2ft \\ &= 2 \left(\frac{\omega}{2\pi} \right) t \\ &= \frac{\omega t}{\pi} \end{aligned} \quad (\text{III.1})$$

$$\begin{aligned} C &= 2ft \\ &= 2 \left(\frac{\omega}{2\pi} \right) t \\ &= \frac{\omega t}{\pi} \end{aligned} \quad (\text{III.1})$$

C is ~~at the~~ number of times ~~of~~ radiation is emitted from a black body. However, ~~let us set the~~ emissivity of the bare electron, e .

When $0 < e < 1$, ~~calculate~~ the energy of T1 ~~is, can be calculated as~~

$$\begin{aligned} \lim_{t \rightarrow \infty} E_{e1} &= \lim_{t \rightarrow \infty} e^C T_e \cos^4 \left(\frac{\omega t}{2} \right) \\ &= \lim_{t \rightarrow \infty} e^{\left(\frac{\omega t}{\pi} \right)} T_e \cos^4 \left(\frac{\omega t}{2} \right) \\ &= 0 \end{aligned} \quad (\text{III.2})$$

$$\begin{aligned} \lim_{t \rightarrow \infty} E_{e1} &= \lim_{t \rightarrow \infty} e^C T_e \cos^4 \left(\frac{\omega t}{2} \right) \\ &= \lim_{t \rightarrow \infty} e^{\left(\frac{\omega t}{\pi} \right)} T_e \cos^4 \left(\frac{\omega t}{2} \right) \\ &= 0 \end{aligned} \quad (\text{III.2})$$

This equation means ~~that~~ the energy of ~~the~~ bare electron ~~decreased~~ decreases with every single absorption in proportion ~~as to the~~ value of the emissivity e . The energy of T2 ~~is~~ derived same as above:

$$\begin{aligned} \lim_{t \rightarrow \infty} E_{e2} &= \lim_{t \rightarrow \infty} e^C T_e \sin^4 \left(\frac{\omega t}{2} \right) \\ &= \lim_{t \rightarrow \infty} e^{\left(\frac{\omega t}{\pi} \right)} T_e \sin^4 \left(\frac{\omega t}{2} \right) \\ &= 0 \end{aligned} \quad (\text{III.3})$$

コメントの追加 [Editor20]: Remark: Please clarify as to what does “it” refers to here. Please specify.

$$\begin{aligned}
\lim_{t \rightarrow \infty} E_{e2} &= \lim_{t \rightarrow \infty} \epsilon^C T_e \sin^4 \left(\frac{\omega t}{2} \right) \\
&= \lim_{t \rightarrow \infty} \epsilon^{\left(\frac{\omega t}{\pi} \right)} T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.3}) \\
&= 0
\end{aligned}$$

These two equations show ~~that~~ the two oscillators behave as if they are damped ~~oscillations~~. ~~And these two indicate, as well as~~ that the masses of ~~the two bare electrons will decrease by and by~~ [2.1 \[JP\] 英単語・語句のご確認をお願いします。](#) ~~and finally to be, eventually reaching zero. As considering~~ Considering the isolated system, ~~the~~ total energy of the whole electron, E_e , is unchangeable:

$$E_e = \lim_{t \rightarrow \infty} E_{e1} + \lim_{t \rightarrow \infty} E_{e2} + \gamma_{K,E}^* \quad (\text{III.4})$$

$$E_e = \lim_{t \rightarrow \infty} E_{e1} + \lim_{t \rightarrow \infty} E_{e2} + \gamma_{K,E}^* \quad (\text{III.4})$$

so,

$$\begin{aligned}
\gamma_{K,E}^* &= E_e - \left(\lim_{t \rightarrow \infty} E_{e1} + \lim_{t \rightarrow \infty} E_{e2} \right) \quad (\text{III.5}) \\
&= E_e
\end{aligned}$$

$$\gamma_{K,E}^* = E_e - \left(\lim_{t \rightarrow \infty} E_{e1} + \lim_{t \rightarrow \infty} E_{e2} \right) \quad (\text{III.5})$$

~~Eq. (III.5)~~ $= E_e$

~~Equation (III.5) concludes~~indicates that the final outcome is only going to be ~~only~~ a photon. ~~Because~~ the total energy of the virtual photon, γ_K ~~occupy, ultimately occupies~~ the total energy of ~~an~~the electron E_e ~~at last~~. In ~~another word~~other words, the phenomenon ~~that~~whereby two oscillators vanish without any positive electronpositrons and change ~~its whole~~the electron's full mass-energy into a real photon ~~conflict~~conflicts with charge conservation. Consequently, the emissivity has to be 1 ~~to meet the law for being as it is~~charge conservation to be maintained.

B. ~~The~~ Virtual Photon as Perfect Fluid

This electron model regards the virtual photon as ~~the~~a perfect fluid. ~~That would have been in, with~~ reference to the

hydrostatic equilibrium and the polytropic equation. ~~The polytropic equation treats the fluid as a perfect fluid.~~

If the virtual photon of the electron had viscosity, the electron could not exist continuously ~~due to~~ because of damped oscillation.

As far as the moving behavior, this condition is equal to the under-damped spring-mass system with $Z \leq 1$, where Z is the damping ratio.

For example, suppose

$$\zeta = e^{-t} \quad (\text{III.6})$$

$$\zeta = e^{-t} \quad (\text{III.6})$$

then the oscillation of the virtual photon is

$$\lim_{t \rightarrow \infty} \zeta \gamma_{K.E.}^* = \lim_{t \rightarrow \infty} e^{-t} \gamma_{K.E.}^* = 0 \quad (\text{III.7})$$

$$\lim_{t \rightarrow \infty} \zeta \gamma_{K.E.}^* = \lim_{t \rightarrow \infty} e^{-t} \gamma_{K.E.}^* = 0 \quad (\text{III.7})$$

Think

~~Were the photon as a composed of viscous fluid, the photon causes it would cause viscous flow and change, changing its kinetic energy into the heat energy. Finally, the photon would cease to oscillate.~~

In this paper, the category "perfect fluid ~~covers the whole fluid which has no~~ comprises all fluids that lack viscosity. That is, both a-compressible fluid and an-incompressible fluids can be included in. One perfect. Were we to treat only the incompressible fluid, that makes inconsistency fluids, this would lead to an inconsistency.

~~It is essential for interaction~~ Interaction between the heat energy and the kinetic energy is essential in this electron model.

~~Because~~ the interaction between the two kinds of energy drives the virtual photon along with the harmonic oscillator. **This**

condition that the harmonics cease to oscillate would be able to exist and need a future study. [12.1 JPL 英単語・語句の確認を](#)

[お願いします。](#)

~~By~~ Moreover, no conditions on the way, it couldn't show any condition about pressure pressure of the virtual photon could be shown in this paper. Consider the compressible For an incompressible fluid, it should be mentioned the pressure of the virtual photon p_{Y^*} in thermodynamics ($p = p(p, T)$) is

$$p_{\gamma^*} = p(\rho_{\gamma^*}, T_{e1}, T_{e2}) \quad (\text{III.8})$$

$$p_{\gamma^*} = p(\rho_{\gamma^*}, T_{e1}, T_{e2}) \quad (\text{III.8})$$

~~Also, any equation relates to an electron comparing teno equations describing the hydrostatic equilibrium cannot be of the electron are shown here. These are; this too is left to future tasks as well work.~~

C. — Interpretation and Adjustment of the Amplitude

コメントの追加 [Editor21]: Remark: We infer that you mean to explain why it cannot be a viscous fluid. Please check.

コメントの追加 [Editor22]: Remark: We infer that you are referring to hydrostatic-equilibrium equations for the electrons. Please check.

In this model, ~~there is a relationship between the amplitude~~the amplitude and the radius of the virtual photon ~~through~~are related by the equations ~~we got that I have determined~~. One of the ambiguous points in this paper is ~~an~~the interpretation and definition of the amplitudes. There are two amplitudes in here. ~~The two appears,~~ appearing in q_e and q_{Y^*} . ~~These amplitudes~~We shall ~~be interpreted cautiously~~interpret them cautiously.

In this subsection, there cannot be a certain conclusion at that point [2.2 [JP] 主語が不明確です]. Instead ~~of that,~~ show, some possibilities ~~effor~~ choosing oscillations ~~at the beginning of~~when making the model ~~are shown~~. How the one is chosen and laid aside the others [2.1 [JP] 英単語・語句のご確認をお願いします。].

Eventually, two possibilities appeared in the scene of taking the harmonics equation in mathematically [2.1 [JP] 英単語・語句のご確認をお願いします。]. There are the two scenarios. ~~For reference~~As a first scenario, set ~~each~~ oscillator as Eqs (III.9a)

and (III.9b), which leads to Eqs (III.9c) and (III.9d)

Eq. (III.9a) and (III.9b), then it leads to Eq. (III.9e)

and (III.9d). Here is a sequence of them,

$$T_{e1} = T_e \cos^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.9a})$$

$$T_{e2} = T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.9b})$$

$$q_e = -T_e \kappa_e \omega \sin(\omega t) \quad (\text{III.9c})$$

$$q_{Y^*} = -A \omega \sin(\omega t) \quad (\text{III.9d})$$

$$(A = T_e \kappa_{Y^*}) \quad (\text{III.9e})$$

The second scenario is, ~~set the oscillator as Eq.~~

(III.10a) and (III.10b), then it gets Eq. (III.10c) and (III.10d).

$$T_{e1}^4 = \left(T_e \cos \left(\frac{\omega t}{2} \right) \right)^4 \quad (\text{III.10a})$$

$$T_{e2}^4 = \left(T_e \sin \left(\frac{\omega t}{2} \right) \right)^4 \quad (\text{III.10b})$$

$$q_e = -T_e^4 \kappa_e \omega \sin(\omega t) \quad (\text{III.10c})$$

$$q_{Y^*} = -A \omega \sin(\omega t) \quad (\text{III.10d})$$

$$(A = T_e^4 \kappa_{Y^*}) \quad (\text{III.10e})$$

$$T_{e1} = T_e \cos^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.9a})$$

$$T_{e2} = T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.9b})$$

$$q_e = -T_e \kappa_e \omega \sin(\omega t) \quad (\text{III.9c})$$

$$q_{Y^*} = -A \omega \sin(\omega t) \quad (\text{III.9d})$$

$$(A = T_e \kappa_{Y^*}) \quad (\text{III.9e})$$

By the second scenario, set the oscillators as Eqs (III.10a) and (III.10b), and then we obtain Eqs (III.10c) and (III.10d):

$$T_{e1}^4 = \left(T_e \cos \left(\frac{\omega t}{2} \right) \right)^4 \quad (\text{III.10a})$$

$$T_{e2}^4 = \left(T_e \sin \left(\frac{\omega t}{2} \right) \right)^4 \quad (\text{III.10b})$$

$$q_e = -T_e^4 \kappa_e \omega \sin(\omega t) \quad (\text{III.10c})$$

$$q_{\gamma*} = -A \omega \sin(\omega t) \quad (\text{III.10d})$$

$$(A = T_e^4 \kappa_{\gamma*}) \quad (\text{III.10e})$$

Whichever ~~we chose~~ scenario ~~one or two~~ we chose, the virtual photon oscillates along with the sine function, q_{Y*} , ~~Eq. (III.9d, III.10d)~~ Eqs (III.9d) and (III.10d).

~~By the way,~~ The total power per unit area, in units of (W/m^2) , is proportional to the fourth power of the absolute temperature. ~~Known with the equation including proportionality between them given by the~~ Stefan—Boltzmann constant, σ_{SB} ,

~~$$\mathcal{E}_{\text{per unit area}} = \sigma_{SB} T^4 \quad (\text{III.11})$$~~

~~Based on the fact~~ $\mathcal{E}_{\text{per unit area}} = \sigma_{SB} T^4 \quad (\text{III.11})$

Thus, this second scenario is useful for thinking about black body radiation, $Q = \sigma_{SB} T^4$, as well as the effective radiation between two things, $Q = \sigma_{SB} (T_1^4 - T_2^4)$, through unifying concept of quadruple T.

~~Actually, According to this model interprets that the way of transfer,~~ heat ~~energy is transferred~~ between the two bare ~~electron is~~ radiation. ~~The~~ electrons by radiation, which is possible ~~way~~ because those two are within the virtual photon, which works as a medium.

One ~~of the~~ major difference comes from the way of ~~having a view of~~ viewing the temperature gradient: ~~For example, the way of setting amplitude A. That difference generates a variation of Eq. (III.9d) and (III.10d).~~

~~For~~ conduction and convection, energy transfer between two objects depends approximately on the temperature difference between them. For natural (free) convection, or when variable property effects are included,

コメントの追加 [A23]: It is not clear how this constitutes a different way of seeing the temperature gradient. Please revise for clarity.

the energy transferred can be a function of the difference between $(T_1)^n$ and $(T_2)^n$, where the power n may be slightly larger than one, and can reach two. Thermal radiation transfer that takes place between two distant bodies depends on the difference between the fourth power of their absolute temperatures ($n = 4$) [4].

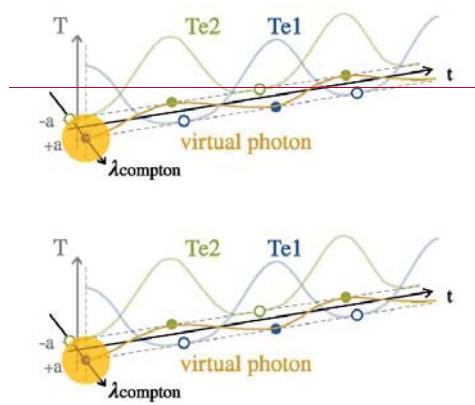


FIG._10. The third scenario. ~~Changed the~~The x-axis ~~has been changed~~ into λ_{Compton} for ~~keeping consistency due to~~ ~~because of the~~ use of $k'y^*$ instead of ky^* . The constant value ky^* is ~~made adjustment~~adjusted by multiplying the Compton wavelength.

This paper had picked up scenario one as ~~Eq. (III.9a)~~Eqs (III.9a) and (III.9b) the first time. The modeling is moving forward on the scenario [2.1 [JP] 英単語・語句のご確認をお願いします。]. ~~The reason This is that the way of~~because energy transfer between the bare electrons by the virtual photon should ~~not~~be considered by ~~radiation but by~~heat conduction, ~~rather than radiation~~. It is certain that a real photon carries ~~the energy of~~as electromagnetic waves, but the virtual photon does not move between the ~~two black bodies~~ at the speed of light. The velocity of ~~a~~ real photon captured is variable. ~~Yet, however~~, energy transfer by radiation travels at the speed of light. There is ~~a~~room for applying the concept of heat conduction, ~~but~~ not radiation, to this model.

However, both ~~scenario has~~scenarios involve a collision ~~in a~~case. The higher the ~~energy of the virtual photon~~photon's energy, the wider its radius due to increasing T_e . That is conflict the fact of behavior of the electron.

~~Alternative~~An alternative approach is ~~putting to somehow put~~ the Compton ~~wavelength~~~~Compton wavelength~~ λ_{Compton} into the Eq. ~~(III.9a) somehow~~Eq. (III.9a). The

Compton wavelength is given by;

$$\lambda_{\text{Compton}} = \frac{h}{mc} \quad (\text{III.12})$$

So,

$$r_{\text{virtual photon}} \propto \frac{h}{T_e c} \quad (\text{III.13})$$

$$\lambda_{\text{compton}} = \frac{h}{mc} \quad (\text{III.12})$$

so

$$r_{\text{virtual photon}} \propto \frac{h}{T_e c} \quad (\text{III.13})$$

It would be preferable for this model on the point of determination of radius of the virtual photon. ~~Because the~~ The higher the electron's energy ~~the electron has~~, the shorter ~~radius of~~ the virtual ~~photon-photon's radius~~ would be. That is, the amplitude A should be defined as the reciprocal of ~~the~~ temperature T.

The third scenario, ~~for example, is to~~ set the oscillators as Eq. (II.14a) Eqs (III.14a) and (II.14b) III.14b). These are the same as in the first scenario ~~one. However, a remarkable point is changed; however,~~ the value of k_7 ! ~~is remarkably changed~~ into as follows, reflecting the Compton wavelength.

$$T_{e1} = T_e \cos^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.14a})$$

$$T_{e2} = T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.14b})$$

$$\mathbf{q}_e = -T_e \kappa'_e \omega \sin(\omega t) \quad (\text{III.14c})$$

$$\mathbf{q}_{\gamma*} = -A \omega \sin(\omega t) \quad (\text{III.14d})$$

$$\left(A = T_e \kappa'_{\gamma*} = \frac{h}{m_e c} \kappa_{\gamma*} \right) \quad (\text{III.14e})$$

$$\left(\kappa'_{\gamma*} = \kappa_{\gamma*} \frac{h}{m_e c} \frac{1}{T_e} \right) \quad (\text{III.14f})$$

$$T_{e1} = T_e \cos^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.14a})$$

$$T_{e2} = T_e \sin^4 \left(\frac{\omega t}{2} \right) \quad (\text{III.14b})$$

$$\mathbf{q}_e = -T_e \kappa'_e \omega \sin(\omega t) \quad (\text{III.14c})$$

$$\mathbf{q}_{\gamma*} = -A \omega \sin(\omega t) \quad (\text{III.14d})$$

$$\left(A = T_e \kappa'_{\gamma*} = \frac{h}{m_e c} \kappa_{\gamma*} \right) \quad (\text{III.14e})$$

$$\left(\kappa'_{\gamma*} = \kappa_{\gamma*} \frac{h}{m_e c} \frac{1}{T_e} \right) \quad (\text{III.14f})$$

T

Then, ~~got Eq. (II.14a) Eqs (III.14c) and (II.14d). See III.14d) can be obtained; see~~ Fig. 10

~~for farther 10 for further information about changed the change from the x-axis to Ae\Compton. In addition, 1/Te in the Eq (II.14f (III.14f) is baseless but for canceling exists only to cancel Te.~~

コメントの追加 [Editor24]: Remark: We infer that you posit its existence to cancel this term. Please check whether this is the case

Finding relationship between [the](#) amplitude and [the](#) radius of the virtual photon for keeping consistency is a challenging task that remains to be solved.

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